Banoth Mukesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.38
TTWRJ COLLEGE BOYS GANDHARI	Class XII (TSBIE)	2022	91.30%
ZPHS BHAVANIPET	Class X (TSBSE)	2020	9.80%

WORK EXPERIENCE

AI Engineer Intern -- Scadea Software Solutions | Hyderabad, India

- **Built FastAPI APIs** for cameras, inspections, workers, and reports within a **layered backend architecture**.
- **Integrated YOLOv8 detections with safety rules** to automate **violation tracking and compliance analysis**.
- **Resolved API integration and duplicate-detection issues** using structured payloads and **worker tracking**.

AI Engineer Intern -- Amani Automation Pvt. Ltd | Hyderabad, India

- Developed a **CNN-based computer vision system** to process the drone's front-view camera feed and classify objects.
- Predicted obstacles in the drone's flight path and sent detection information to the **STM32 microcontroller** for **real-time decision-making** and accident avoidance.

SKILLS

Languages: C, C++ (OOPs, STL), Python, SQL | **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch

Computer Vision: OpenCV, Ultralytics (YOLOv8) | **Core Skills:** Machine Learning, Deep Learning, DSA, Data Science

CS Fundamentals: OS, DBMS | **Coding Skill:** Solved **300+ problems** in Leetcode(**Link**)

PROJECTS

Driver Drowsiness Detection System | [Link]

- Developed a **real-time CNN-based** drowsiness detection system using **2,900 facial images** to identify driver drowsiness.
- Trained the CNN model on facial images, achieving **96% validation accuracy** with a strong F1-score for classification.
- Integrated **Haar Cascade** face and eye detection with video streams and **audio alerts** for consecutive eye closure detection.

Laptop Price Predictor Web Application | Machine Learning Project | [Link]

- Developed a machine learning web app to predict laptop prices using **1,303 records** and features like RAM, CPU, GPU.
- Cleaned and preprocessed the laptop dataset by handling missing values, removing duplicates, and encoding features.
- Compared **Regression, Lasso, Ridge, Random Forest**, and XGBoost models, achieving an **R^2 score of 0.88 with XGBoost**.

AeroInspect AI | AI-Powered Construction Safety Monitoring Platform | [Link]

- Built an AI-powered safety platform using **YOLOv8, FastAPI, and PostgreSQL** for real-time violation monitoring.
- Developed a rule engine to evaluate PPE detections, calculate **compliance scores**, and trigger critical alerts.
- Integrated **LLM-based PDF analysis** to extract safety rules and built dashboards for inspections and reporting.

Custom STM32-Based Flight Controller for UAVs | [Link]

- Developed a custom STM32 flight controller with **MPU6050 IMU** for real-time UAV motion sensing and control.
- Implemented **Kalman-filter sensor fusion** to combine accelerometer and gyroscope data for orientation estimation.
- Designed dual-loop PID control for pitch and roll stabilization with **PPM decoding** for RC transmitter inputs.

RELEVANT COURSES

Computer Science Electives: Introduction to Programming, Introduction to Artificial Intelligence

Mathematics Electives: Calculus I and II, Elementary Linear algebra, Differential Equations, Introduction to Probability

supervised Machine Learning: Regression&Classification--DeepLearning.AI & Stanford University

CERTIFICATE

- **Google Student Upskilling Launchpad** -Completed with dedicated participation and successful attendance.
- **Tata Crucible Campus Quiz 2025** -Participated in Prelims Level 1.

POSITIONS OF RESPONSIBILITY

Hostel Representative (HR), General Council || IITH

- Represented **330 students** to the Dean's Office and **Hostel Committee** to solve daily issues.
- Led the hostel team in **25+ tech and cultural events** in IITH GCC, winning **1st place overall**.
- Worked with the **Student Council** in regular meetings to fix student problems and complaints.

Coordinator - Pronites & Production || Milan IITH

- Worked with a 15-member team to deliver **4 unique Pronites** during IITH's week-long annual sports fest.
- Directed end-to-end artist management; negotiated with **10+ agencies** to finalize a **headline band**.
- Executed **stage production and logistics**, ensuring seamless on-ground delivery of **live performances**.

EXTRACURRICULAR

CPL (College People Leader) -- TTWRJC Boys,Gandhari | kamareddy

- Addressed **student concerns** and communicated their needs to the **principal**, helping implement solutions for all.
- Conducted **daily roll calls and meetings** while supporting **student discipline** and coordinating daily activities.

Member, National Sports Organization (NSO) -- IITH

- Represented IITH as a **volleyball player**, attending regular practice to build **fitness, discipline and teamwork.**
- Helped IITH secure **2nd Runner-Up** in the **Inter-IIT Sports Meet 2025** through consistent training and team unity.